Assignment 5: Matlab MA 122
Course Instructor:SK+MKR
Department of Mathematical Sciences
Rajiv Gandhi Institute of Petroleum Technology Jais Amethi

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1. Given two vectors, x and y. The x values begin at 0 and end at 15π, with a spacing of $\frac{\pi}{100}$, while the y values begin at 0 and end at 12, with a spacing of $\frac{\pi}{10}$.
   (a) Find size, bytes and class of x and y.
   (b) Save input and output using 'diary'.

2. Create a vector x of values from 0 to 10π, with a spacing of $\frac{\pi}{100}$. Define vector y as
$$y = \pi \sin(x)$$
   (a) Find size, bytes and class of x and y.
   (b) Save input and output using 'diary'.

3. Given $y1 = 2\cos(x)$ and $y2 = 5\sin(x)$. Let x vary from 0 to 2π in increments of 0.1π.
   (a) Save and load workspace variables.
   (b) Create new script, load the data saved in the workspace and plot x versus $y1$.
   (c) Create new script, load the data saved in the workspace and plot x versus $y2$.
   (you may use 'doc' command for help)

4. Given $y1 = 2x$ and $y2 = 5\sin(x)$. Let x vary from 0 to π in increments of 0.1π.
   (a) Save and load workspace variables.
   (b) Create new script, load the data saved in the workspace and plot x versus $y1$.
   (c) Create new script, load the data saved in the workspace and plot x versus $y2$.

5. Create a vector x of values from 0 to 2π, with a spacing of $\frac{\pi}{4}$. Define vector y as
$$y = \sin(x)$$
   Next, define a new vector, xq of values from 0 to 2π, with a spacing of $\frac{\pi}{16}$.
   (a) Find the yq values corresponding to the xq values by linear interpolation.
   (b) On the same figure, plot the original y vs. x as circles, and yq vs. xq as a line.
   (c) Repeat the exercise in part (a) and (b) using the spline(...) function to interpolate.

6. Create a vector x of values from 1 to 9. Define vector v as
$$v = [0 \;\; 1.41 \;\; 2 \;\; 1.41 \;\; 0 \;\; -1.41 \;\; -2 \;\; -1.41 \;\; 0]$$
   .
   Next, define a new vector xq of values from 1.5 to 8.5.
   (a) Find the yq values corresponding to the xq values by linear interpolation.
   (b) On the same figure, plot the original y vs. x as circles, and yq vs. xq as a line.
   (c) Repeat the exercise in part (a) and (b) using the spline(...) function to interpolate.